

National University of Computer and Emerging Sciences



Lab Manual 03 Programming Fundamentals

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Section	BCS-1B
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Objectives

- The purpose of this lab is to familiarize you with the programming tool. Lo and behold! You will be working on the **Microsoft Visual Studio** through your journey to learn programming in this course. You will be able to understand how to make a

project in visual studio, how the program is executed and what the sequence of the output is.

- Pseudo Codes (Loops)

Task 1:

Write a pseudo code using loops that prints the numbers from 1 to 100. You should also print the word 'tea' after every even number. Output will be 12tea34tea.....

Task 2:

Write a pseudo code using loops that takes a number N and print the N numbers of the Following sequence: 1, 2, 4, 7, 11, 16, 22, 29... If N is 3, then print 1, 2, and 4.

Task 3:

Write a pseudo code that keep taking input numbers from user and terminate the program when user enters -1. Your task is to determine either user enter any number 3 time in a row or not.

For example

1) 4 5 1 6 3 1 2 1 -1
Return false

2) 5 4 3 1 3 4 1 6 6 6 4 3 1 -1
Return True as 6 entered 3 times in a row.

Task 4:

Write a pseudo code to display the cube of the number upto given an integer.

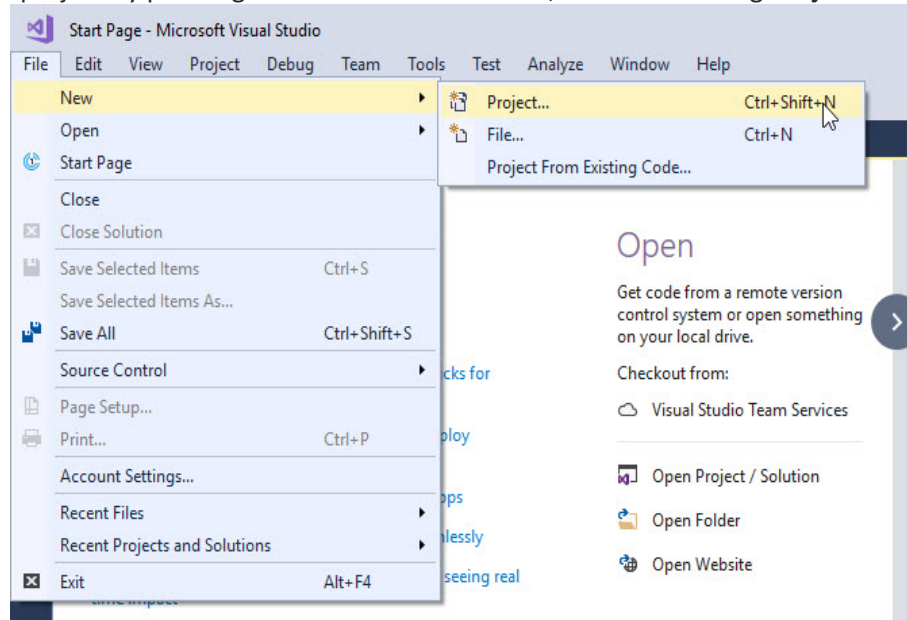
Task 0: My First Computer Program

Creating a New Project

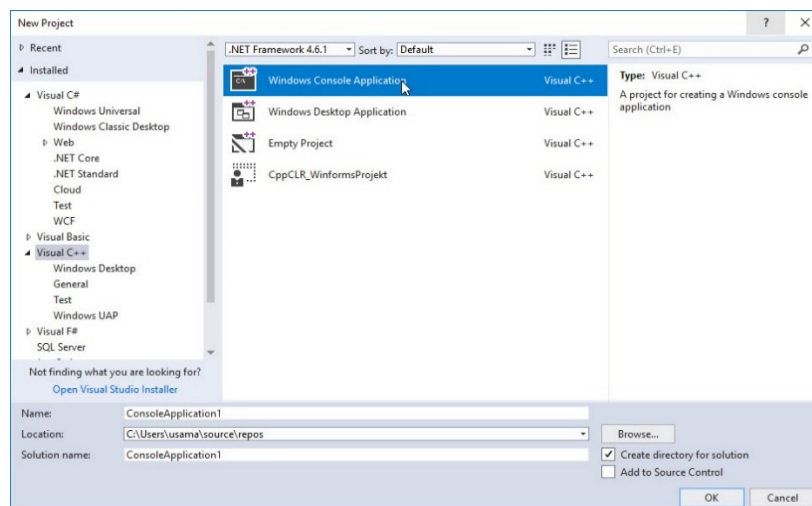
It's time to start! Go ahead and open up Visual Studio from the desktop or start menu.

When you create an app in Visual Studio, you first create a project and a solution. For this example, you'll create a Windows console application.

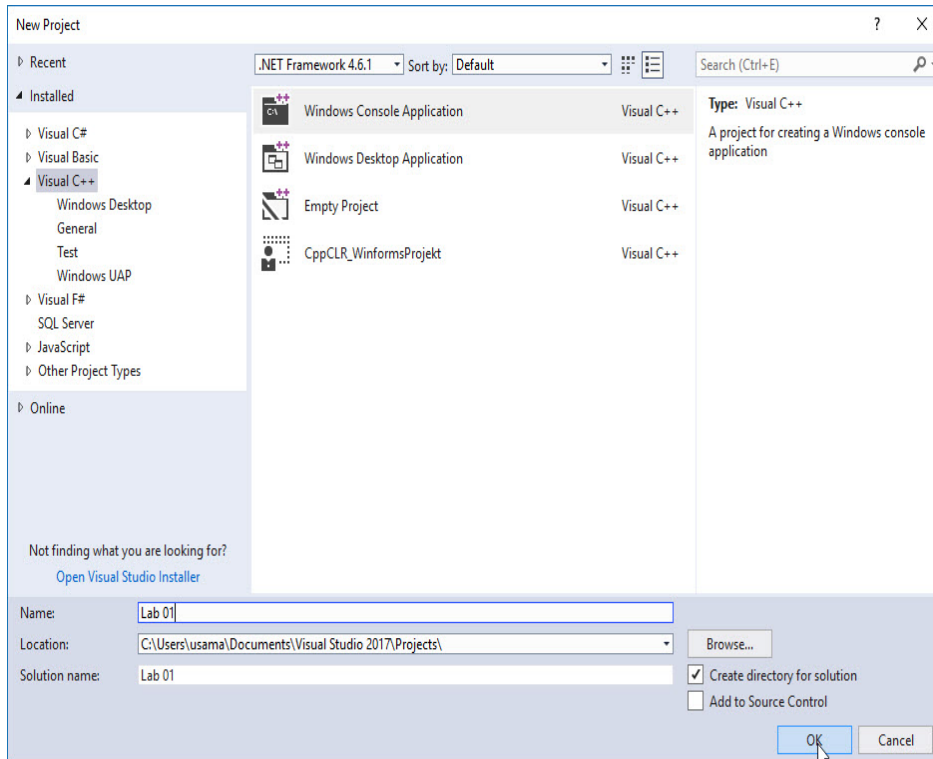
- Create a project by pointing to **New** on the **File** menu, and then clicking **Project**.



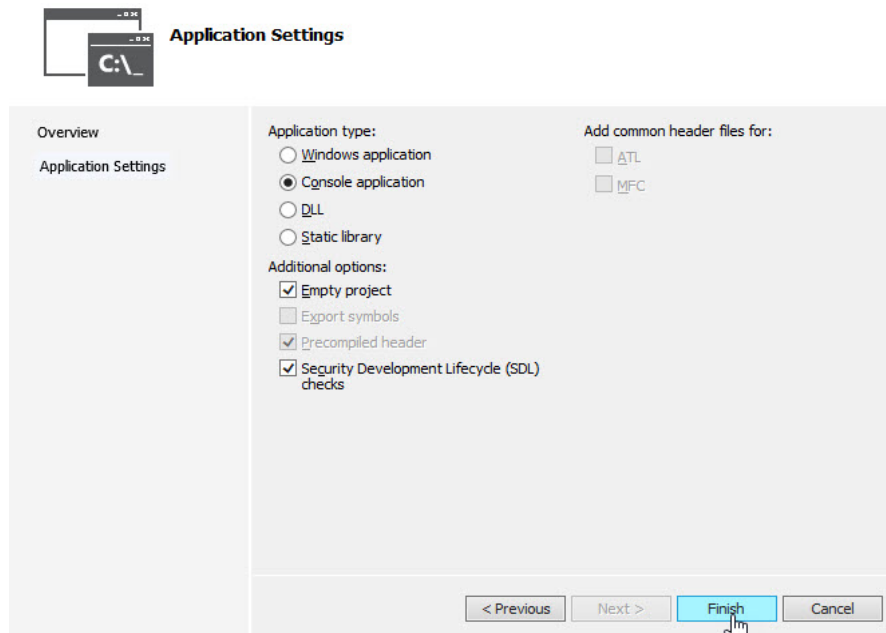
- In the **Visual C++** project types pane, click **Win32**, and then click **Win32 Console Application**.



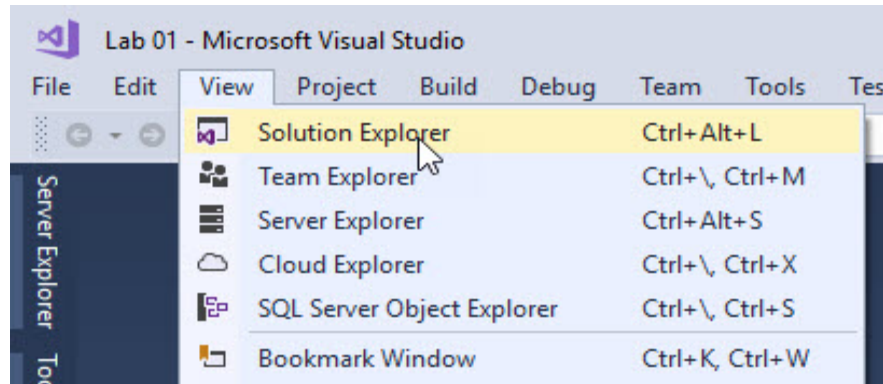
- Type a name for the project.



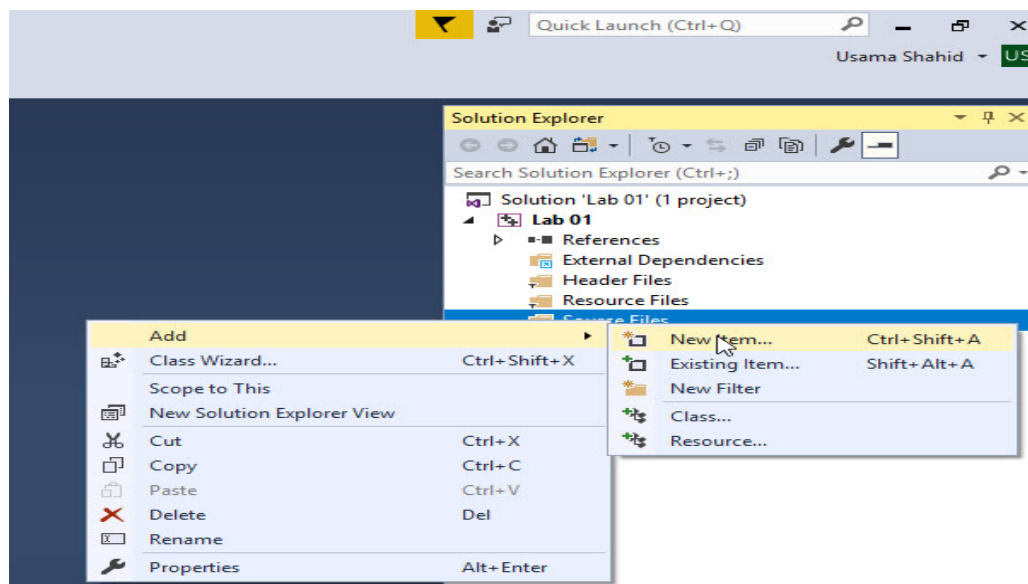
- By default, the solution that contains the project has the same name as the project, but you can type a different name. You can also type a different location for the project.
- Click **OK** to create the project.
- In the **Win32 Application Wizard**, click **Next**, select **Empty Project**, and then click **Finish**.



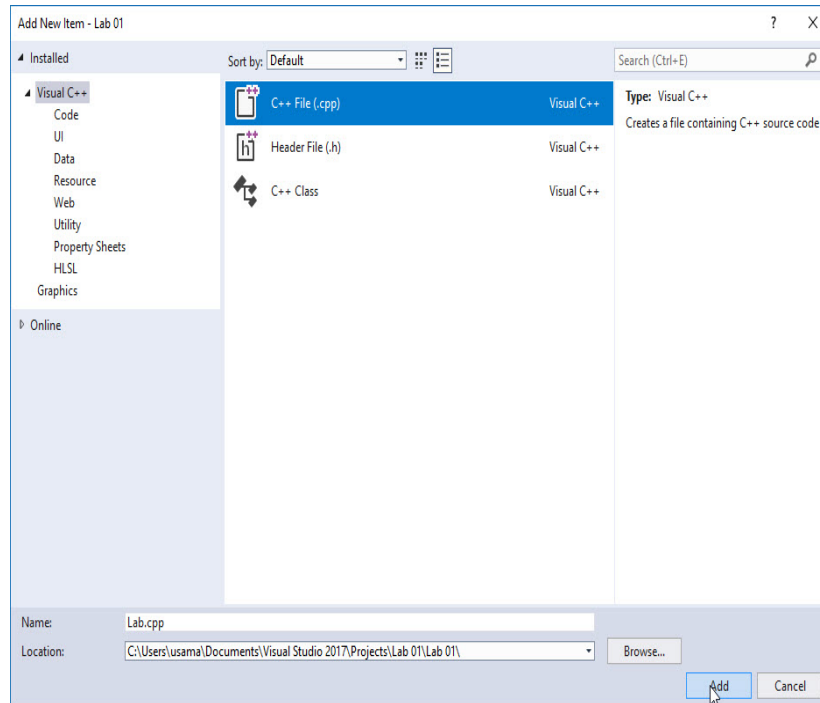
- If **Solution Explorer** is not displayed, on the **View** menu, click **Solution Explorer**.



- Add a new source file to the project, as follows.
 - In **Solution Explorer**, right-click the **Source Files** folder, point to **Add**, and then click **New Item**.



- In the **Visual C++** node, click **C++ File (.cpp)**, type a name for the file, and then click **Add**.



- The **.cpp** file appears in the Source Files folder in **Solution Explorer**, and the file is opened in the Visual Studio editor.
- In the file in the editor, type a valid C++ program.

➤ Task 1:

```

Lab 1.cpp
1  #include <iostream>
2  using namespace std;
3
4  int main()
5  {
6
7      cout<<"Programming";
8      cout<<"is great";
9      cout<<"fun :D" << endl;
10
11     return 0;
12
13 }

```

- Write the above code in the file in the editor.
- Compile the code and observe the program output.

➤ **Task 2:**

Write a c++ program to print **“Hello!! Welcome in the world of programming”** on console.

➤ **Task 3:**

Write a C++ program to print “your name, roll no., degree, section and lab Number” in the following format.

Welcome to Programming Fundamentals

Name: Sonia Anum

Roll #: Mscs19044

Degree: MS (CS)

Section: A

Lab: 03
